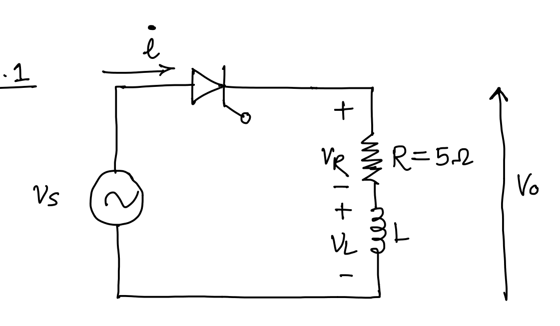
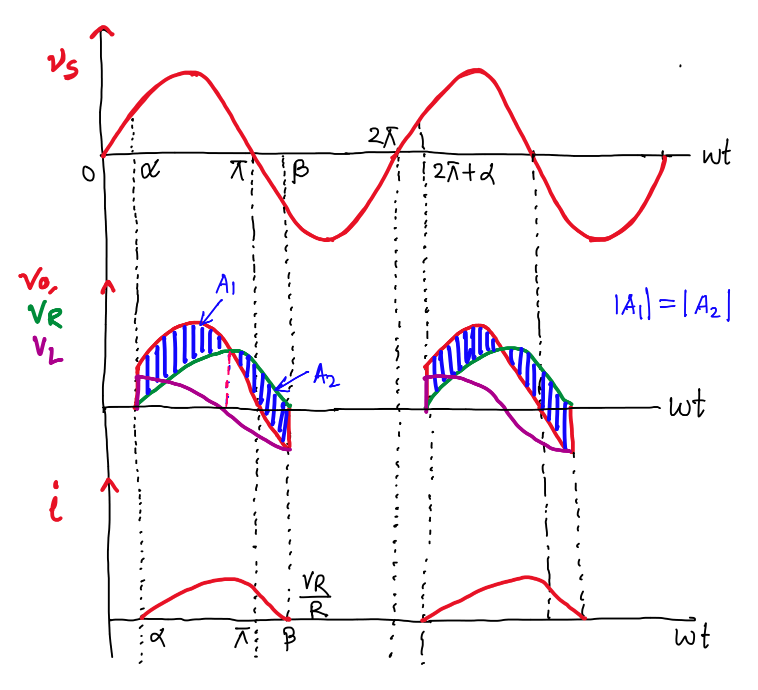
Tuesday, 7 March 2023 2:34 PM

Tutorial 3 Solution Sheet

Q.1



$V_s = 230\sqrt{2}\,\sin(2\pi 50 t)$



Average output voltage,

$$V_o = \frac{1}{2\pi} \times 230\sqrt{2}\left[\cos\alpha - \cos\beta\right]$$

$\alpha = 45^\circ$, $\beta = 200^\circ$

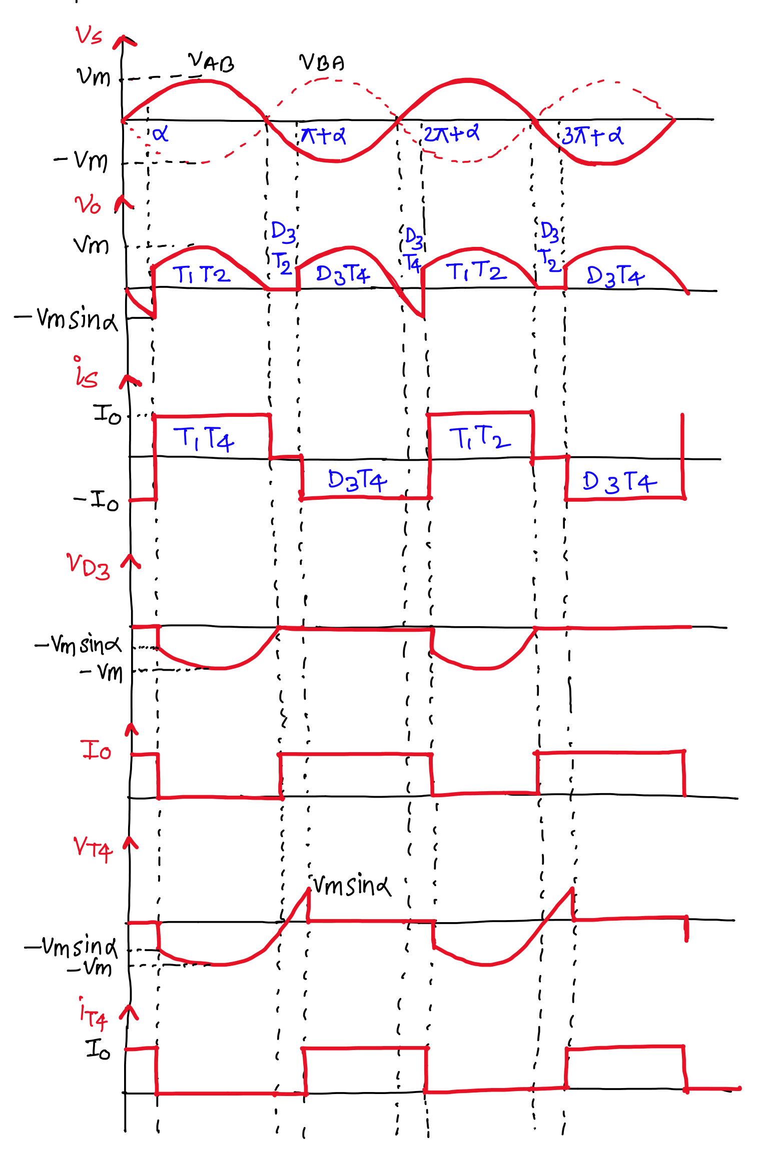
$\boxed{\therefore\ V_o = 85.252 \text{ volts}}$

Circuit turn-off time, $t_c = (360^\circ - \beta) \times \frac{20}{360}$ ms

$\boxed{\therefore\ t_c = 18.89 \text{ ms}}$

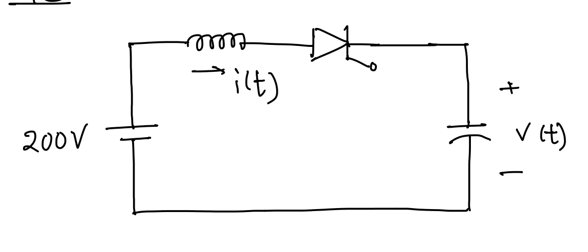
Average load current, $I_o = \frac{V_o}{R} = 17.0504$ A

Q.2

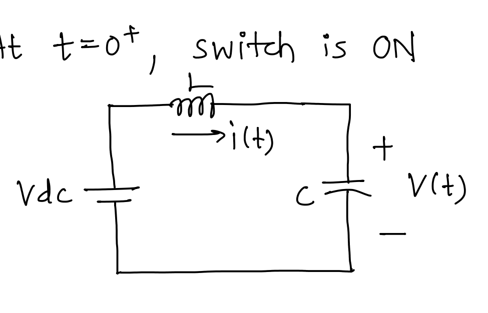


$$V_o(avg) = \frac{V_m}{2\pi}\left\{1 + 3\cos\alpha\right\}$$

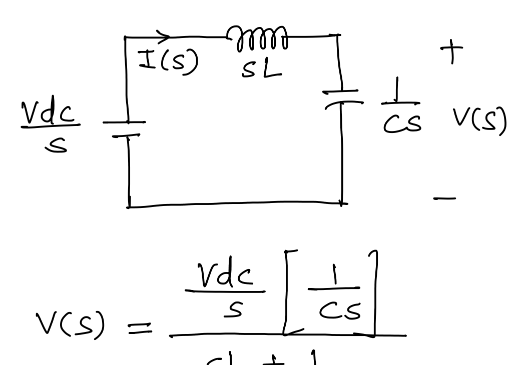
Q.3



At $t = 0^+$, switch is ON



Converting to s-domain



$$V(s) = \frac{\frac{Vdc}{s}\left[\frac{1}{Cs}\right]}{sL + \frac{1}{Cs}}$$

$$= \frac{Vdc}{s\left[s^2LC + 1\right]}$$

$$= \frac{Vdc}{sLC\left[s^2 + \frac{1}{LC}\right]}$$

$$sV(s) = \frac{Vdc}{LC\left[s^2 + \frac{1}{LC}\right]}$$

$$= \frac{Vdc\left(1/\sqrt{LC}\right)}{LC \cdot \frac{1}{\sqrt{LC}}\left(s^2 + \left(\frac{1}{\sqrt{LC}}\right)^2\right)}$$

$$\frac{dV}{dt} = \frac{Vdc}{\sqrt{LC}} \cdot \sin w_o t \quad , \quad w_o = \frac{1}{\sqrt{LC}}$$

$$\therefore\ V(t) = \frac{Vdc}{\sqrt{LC}} \int_0^t \sin w_o t \; dt$$

$$= \frac{-Vdc}{\frac{1}{\sqrt{LC}} \cdot \sqrt{LC}}\left[\cos w_o t - 1\right] + k$$

$$V(t) = Vdc\left[1 - \cos w_o t\right] + k$$

At $t = 0$, $V(t) = 0$

$\therefore\ k = 0$

$$\therefore\ \boxed{V(t) = Vdc\left[1 - \cos w_o t\right]}$$

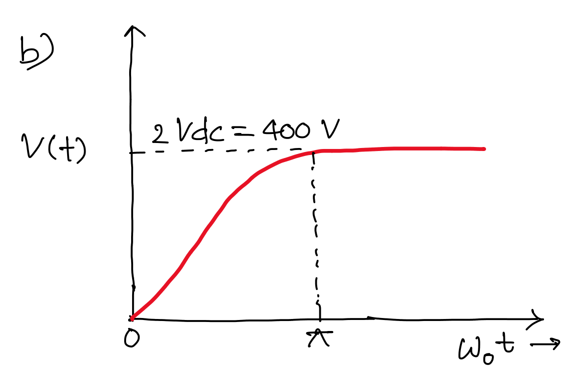
$$\therefore\ I(s) = \frac{Vdc/s}{sL + \frac{1}{Cs}} = \frac{C\,Vdc}{s^2LC + 1}$$

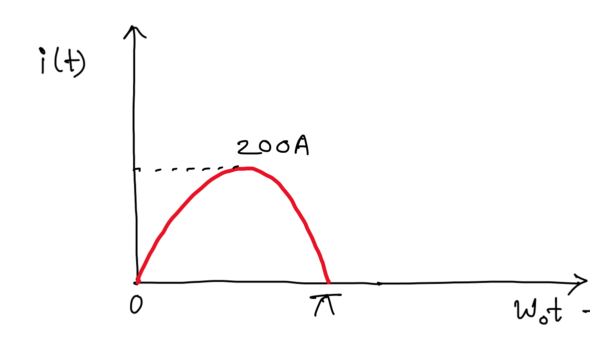
$$I(s) = \frac{C\,Vdc}{s^2LC + 1} = \frac{C}{LC}\,\frac{Vdc}{\left(s^2 + \frac{1}{LC}\right)}$$

$$\therefore\ I(s) = \frac{Vdc \times \frac{1}{\sqrt{LC}}}{\frac{1}{\sqrt{LC}} \cdot L \cdot \left(s^2 + \left(\frac{1}{\sqrt{LC}}\right)^2\right)}$$

$$\boxed{\therefore\ i(t) = Vdc\sqrt{\frac{C}{L}}\,\sin w_o t \ ; \ w_o = \frac{1}{\sqrt{LC}}}$$

b)





c) $E_L = \frac{1}{2} L i^2 = 0$

$E_{cap} = \frac{1}{2} CV^2 = \frac{1}{2} \times 20\mu \times 400^2 = 1.6$J

d) $(E_L)_{peak} = \frac{1}{2} L i_{pk}^2 = \frac{1}{2}(20\mu)(200)^2 = 0.4$J

$(E_{cap})_{peak} = 1.6$J

e) $E_{source} = \left(\int Vdc\ i(t)\ dt\right)$

$$= \left(\int_0^\pi 200\,(200) \sin w_o t \; d(w_o t)\right) / w_o$$

$$= \left(200 \times 200 \times \left[-\cos w_o t\right]_0^\pi\right) / w_o$$

$$= \frac{200 \times 200 \times 2}{W_o} = 200 \times 200 \times 2 \times 20\mu$$

$$= 1.6\text{J}$$

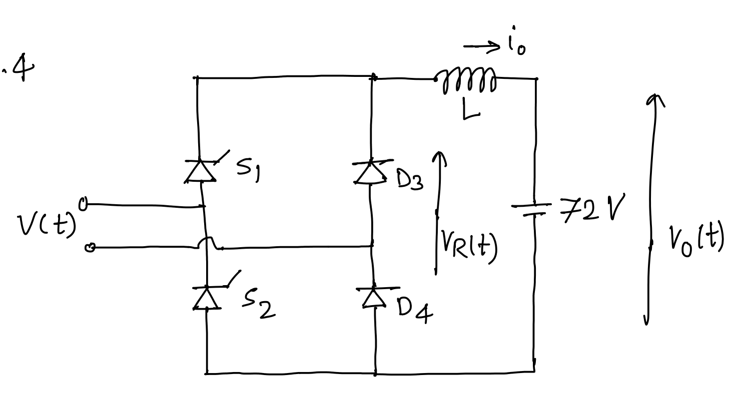
$\therefore$ Energy balance : $E_{source} = 1.6$J

$\therefore\ E_L + E_C = 0 + 1.6\text{J} = 1.6$J

f) Peak Conducting current stress $= I_{pk} = 200$A

Peak blocking voltage stress $= 200$ V

Q.4



$\Delta I_o = 0.1 I_o$

a) $\theta = \sin^{-1}\left(\frac{72}{V_m}\right) = \sin^{-1}\left(\frac{72}{110\sqrt{2}}\right)$

$\theta = 27.57^\circ$

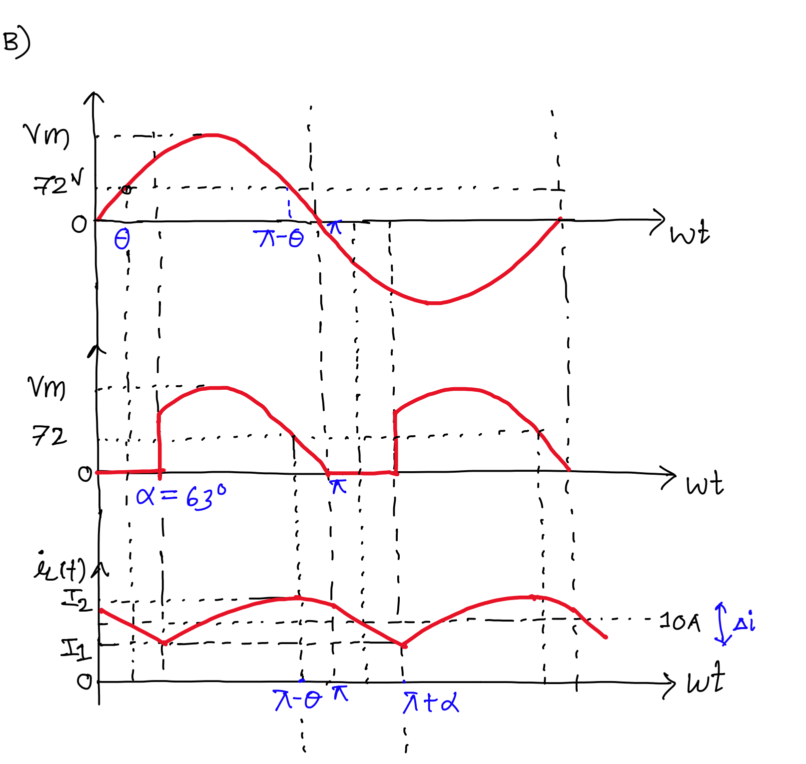
$\therefore\ (V_R)_{avg} = 72$ V

$\therefore\ 72 = \frac{V_m}{\pi}\left[1 + \cos\alpha\right]$

$\frac{72 \times \pi}{110\sqrt{2}} - 1 = \cos\alpha$

$\boxed{\therefore\ \alpha = 63^\circ}$

B)



c)

The current reaches maximum at 'π−θ' and minimum at π+α.

From π−θ to π, the voltage across inductor is

$$v_L = V_m \sin\omega t - E.$$

From π to π+α, the freewheeling happens. & the voltage across inductor is

$$v_L = -E$$

$$v_L = L\frac{di_o}{dt}$$

$$\Rightarrow v_L\,dt = L\,di_o$$

$$\Rightarrow V_L\,d\omega t = \omega L\,di_o$$

$$\Rightarrow \int_{\pi-\theta}^{\pi+\alpha} V_L\,d\omega t = \int_{I_2}^{I_1} \omega L\,di_o$$

$$\Rightarrow \int_{\pi-\theta}^{\pi} (V_m \sin\omega t - E)\,d\omega t + \int_{\pi}^{\pi+\alpha} -E\,d\omega t = \omega L(I_1 - I_2)$$

$$\Rightarrow V_m(-\cos\omega t)\Big|_{\pi-\theta}^{\pi} - E(\omega t)\Big|_{\pi-\theta}^{\pi} - E(\omega t)\Big|_{\pi}^{\pi+\alpha} = -\omega L \Delta I_o$$

$$\Rightarrow V_m(1 - \cos\theta) - E(\alpha + \theta) = -\omega L \Delta I_o$$

$$\Rightarrow L = \frac{E(\alpha+\theta) - V_m(1-\cos\theta)}{\omega \cdot \Delta I_o}$$

$$\Rightarrow L = \frac{72\left(63^\circ + 27.57^\circ\right) \times \frac{\pi}{180^\circ} - 110\sqrt{2}\left(1 - \cos 27.57^\circ\right)}{2\pi \times 50 \times 1}$$

$$\therefore\ L = 306.05 \text{ mH}$$